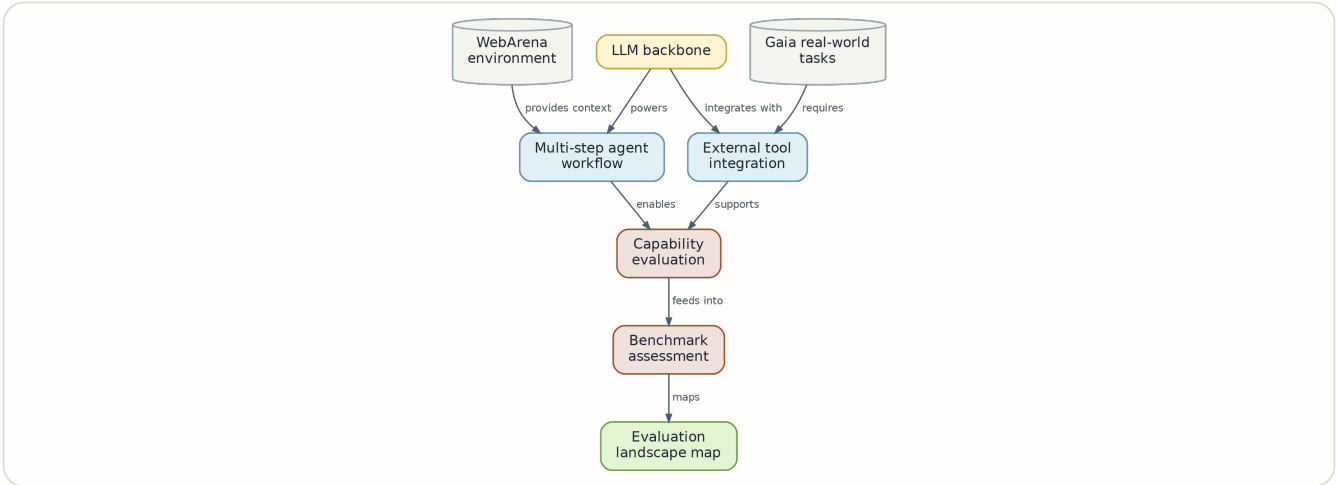


Survey on Evaluation of LLM-based Agents

Asaf Yehudai, Lilach Eden, Alan Li, Guy Uziel, Yilun Zhao, Roy Bar-Haim, Arman Cohan, Michal Shmueli-Scheuer

<https://arxiv.org/abs/2503.16416>



Abstract

LLM-based agents represent a paradigm shift in AI, enabling autonomous systems to plan, reason, and use tools while interacting with dynamic environments. This paper provides the first comprehensive survey of evaluation methods for these increasingly capable agents. We analyze the field of agent evaluation across five perspectives: (1) Core LLM capabilities needed for agentic workflows, like planning, and tool use; (2) Application-specific benchmarks such as web and SWE agents; (3) Evaluation of generalist agents; (4) Analysis of agent benchmarks' core dimensions; and (5) Evaluation frameworks and tools for agent developers. Our analysis reveals current trends, including a shift toward more realistic, challenging evaluations with continuously updated benchmarks. We also identify critical gaps that future research must address, particularly in assessing cost-efficiency, safety, and robustness, and in developing fine-grained, scalable evaluation methods.

Task

Mapping LLM Agent Evaluation Landscape

- LLM-based agents integrate models into multi-step workflows with external tools for autonomous planning.
- This survey presents the first overview of LLM-based agent evaluation by mapping the current landscape.
- The survey covers fundamental capabilities like planning, tool use, self-reflection, and memory.
- It reviews benchmarks for web, software engineering, scientific, and conversational agents.

Challenges

Why Evaluating Agents Is Hard

- Evaluation must go beyond textual outputs to assess sequential decision-making in dynamic environments.
- Benchmarks reveal current methods struggle with long-range consistency and dynamic memory handling.
- Recent work exhibits over-optimistic estimates, proposing Online-Mind2Web as a more rigorous alternative.
- Most benchmarks conflate LLM capabilities with agent harness design, obscuring performance attribution.

Proposed method

Benchmarks: WebArena, SWE-bench, Gaia

- WebArena provides dynamic environments with fully functional websites across multiple domains.
- Gaia requires reasoning, multi-modality, web browsing, and tool-use on real-world questions.

Main results

Performance Gaps and Critical Limitations

- State-of-the-art models struggle with long-horizon planning tasks.
- SWE-bench Pro performance remains below 25% Pass@1 for complex code changes.
- No standardized benchmark exists for assessing self-reflection capabilities.
- Evaluations overlook cost, creating resource-intensive agents hard to deploy.